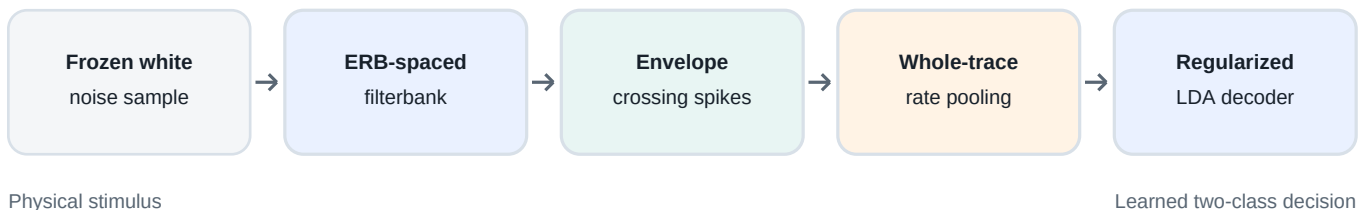


Frozen-Noise Discrimination

A cochlea-like rate-code simulation of why short frozen white-noise samples can remain perceptually distinct while longer samples become statistically alike

Central results. Fixed readout noise exposes the loss of frozen-exemplar identity with duration. At 20 spikes/s noise, same-source exemplar accuracy falls from 99.8% at 25 ms to 70.9% at 1.6 s, while a subtle white versus light-pink source contrast rises from 86.8% to 99.8%. Chance is 50%.

Model at a glance



Purpose

This project is a compact computational test of one explanatory idea: short random waveforms carry accidental, sample-specific patterns across auditory frequency channels; those fingerprints shrink as the sample is averaged for longer; and a noisy cortical readout eventually cannot learn the difference between two frozen samples.

Current scope

- The encoder is cochlea-like rather than a detailed auditory-nerve model.
- The decoder retains average channel rates and discards precise spike timing.
- The reported sweep manipulates fixed Gaussian readout noise after spike-rate pooling.
- All results are Monte Carlo estimates from 12 independently generated frozen-noise pairs.

Project version 0.2 | NumPy-only implementation | Reproducible seeds: 7 and 29

1. Question and modeling hypothesis

The phenomenon

Repeated presentations of a short, fixed realization of white noise can sound like a recognizable object: listeners can learn that frozen waveform and distinguish it from another frozen waveform. As duration increases, the samples' empirical statistics converge toward the same white-noise distribution, and the distinctiveness is expected to weaken.

The proposed mechanism

A finite white-noise sample is not perfectly flat in every auditory band. Short samples have large random deviations in bandwise envelope events. Longer samples average those deviations away. A discriminator operating on noisy, time-averaged channel rates therefore loses class information with duration.

Three linked hypotheses

Hypothesis	Model implication	Observable result
H1 - finite-sample fingerprints	Two frozen samples create different threshold-crossing rates across cochlear channels.	The clean channel-rate distance is positive for every tested duration.
H2 - convergence with duration	Whole-trace averaging makes accidental bandwise deviations smaller for long samples.	The clean rate distance falls as duration grows.
H3 - noise-limited learning	Fixed post-pooling readout noise does not shrink with the sample fingerprint.	Accuracy approaches chance sooner at higher readout-noise levels.

Why a rate code matters

The present decoder receives one feature per cochlear channel: the number of threshold-crossing spikes divided by the sample duration. This intentionally creates a low-dimensional summary of each sound. It is the central modeling assumption, not a neutral preprocessing choice.

If the decoder instead receives the complete time-locked spike sequence, a longer frozen trace contains more identifying events and may become easier to classify. Comparing rate and temporal codes is therefore the most diagnostic next experiment.

What would count as support?

- A monotonic reduction in sample-specific clean feature distance with duration.
- A clear interaction in which readout noise has little effect on very short samples but progressively corrupts long-sample classification.
- Results that replicate across independently generated pairs rather than depending on one unusually distinctive waveform.

2. Model architecture

The implementation uses transparent components and only NumPy. It is fast enough for parameter sweeps while keeping each transformation inspectable.

Stage	Implementation	Function in the hypothesis
Stimulus	Two independent Gaussian white-noise waveforms per pair; each waveform is demeaned and normalized to equal RMS.	Creates repeatable frozen exemplars while removing the trivial overall-level cue.
Cochlear filterbank	16 overlapping fourth-order frequency responses with center frequencies equally spaced on the ERB-number scale from 200 to 7,000 Hz.	Turns accidental spectral fluctuations into a channel population code.
Inner-hair-cell proxy	The analytic envelope is computed in every channel. Upward crossings of a fixed threshold generate Boolean spikes, with a 1 ms refractory interval.	Produces sparse events from changing subband envelopes.
Trial noise	Three percent spike dropout and 0.2 spontaneous spikes/s are applied before pooling. Gaussian readout noise is added after pooling.	Separates noise that can average over time from noise that remains at the final decision representation.
Feature code	Spike count per channel divided by sample duration, yielding a 16-dimensional rate vector in spikes/s.	Preserves the channel fingerprint but discards precise spike timing.
Classifier	Two-class linear discriminant analysis with isotropic covariance regularization.	Learns a stable linear boundary from noisy repetitions.

Signal processing details

- The filterbank is applied in the Fourier domain. Each response is normalized to equal expected RMS for ideal white noise, so high-frequency channels do not win solely because their bandwidth is larger.
- A NumPy Hilbert transform supplies the channel envelope. The default crossing threshold is 1.45 in normalized envelope units.
- Stimulus RMS normalization means the classifier must use within-spectrum structure rather than total loudness.
- The clean fingerprint distance is the root-mean-square Euclidean difference between the two class rate vectors across channels.

Default cochlear settings

Parameter	Value	Parameter	Value
Sample rate	16,000 Hz	Channels	16
Frequency range	200-7,000 Hz	Bandwidth scale	1.5 ERB
Envelope threshold	1.45	Refractory interval	1.0 ms
Stimulus normalization	Equal RMS	Feature units	Spikes/s

Interpretive boundary. ERB spacing, envelope crossings, and a refractory interval make the front end auditory-inspired. They do not make it a validated model of cochlear mechanics, auditory-nerve adaptation, or cortical population dynamics.

3. Experiment and evaluation

Protocol

Step	Operation
1	Choose one of seven durations: 25, 50, 100, 200, 400, 800, or 1,600 ms.
2	Generate 12 independent pairs of frozen Gaussian white-noise samples at that duration.
3	Encode both samples once into clean cochlear threshold-crossing counts.
4	Generate 32 noisy training repetitions and 96 noisy test repetitions for each class.
5	Fit regularized LDA to the pooled rate vectors and score held-out two-alternative accuracy.
6	Repeat at four post-pooling readout-noise levels and average accuracy across frozen pairs.

Noise manipulations

Noise source	Default	Where it enters	Expected dependence on duration
Spike dropout	3%	Before rate pooling	Partly averages over more events in longer samples.
Spontaneous spikes	0.2 spikes/s/channel	Before rate pooling	Poisson variability in normalized rate decreases with duration.
Readout noise	0, 10, 20, or 40 spikes/s SD	After rate pooling	Remains fixed while the frozen-sample fingerprint shrinks.

Classifier and uncertainty

For each pair, the classifier is trained and tested only on repetitions of the same two frozen samples. Accuracy therefore measures whether the model can learn sample identity, not whether it can generalize to a new category of noise. Curves show mean held-out accuracy across 12 frozen pairs; error bars are ± 1 standard error of the mean across pairs.

Controls built into the first version

- Equal RMS removes overall loudness as a class cue.
- The stimulus random-number stream is independent of the trial-noise stream, so adding or removing noise conditions does not change the underlying frozen samples.
- Training and test trials are separate, preventing direct evaluation on the fitted examples.
- A fixed random seed makes the supplied result files reproducible.

What the experiment does not yet test

- Acoustic noise mixed with the waveform before the cochlear filterbank.
- Trial-to-trial filter or threshold jitter inside cochlear channels.
- A leaky cortical integrator with an independently varied time constant.
- Recognition after temporal shifts, onset uncertainty, or changes in presentation level.

4. Initial results

Held-out classification accuracy

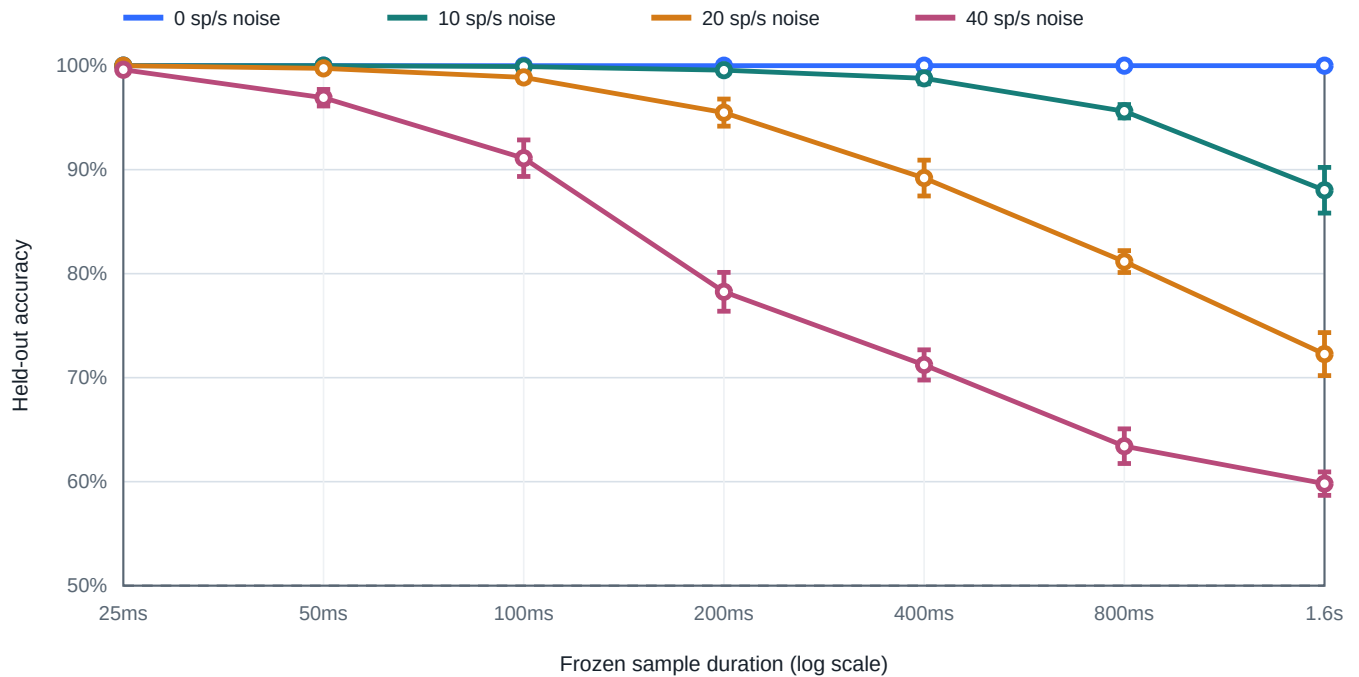


Figure 1. Mean held-out accuracy across 12 frozen pairs. Error bars show ± 1 SEM. The no-readout-noise condition remains at ceiling, while stronger readout noise produces an increasingly steep duration-dependent decline.

Duration	0 sp/s	10 sp/s	20 sp/s	40 sp/s
25 ms	100.0%	100.0%	100.0%	99.6%
100 ms	100.0%	99.9%	98.9%	91.1%
400 ms	100.0%	98.8%	89.2%	71.2%
800 ms	100.0%	95.6%	81.2%	63.4%
1.6 s	100.0%	88.0%	72.3%	59.8%

Main pattern. At 25 ms, all four conditions are at or above 99.6% accuracy. By 1.6 s, accuracy has separated sharply by readout noise: 100.0%, 88.0%, 72.3%, and 59.8% for 0, 10, 20, and 40 spikes/s respectively.

5. Mechanistic interpretation

The frozen-sample fingerprint shrinks

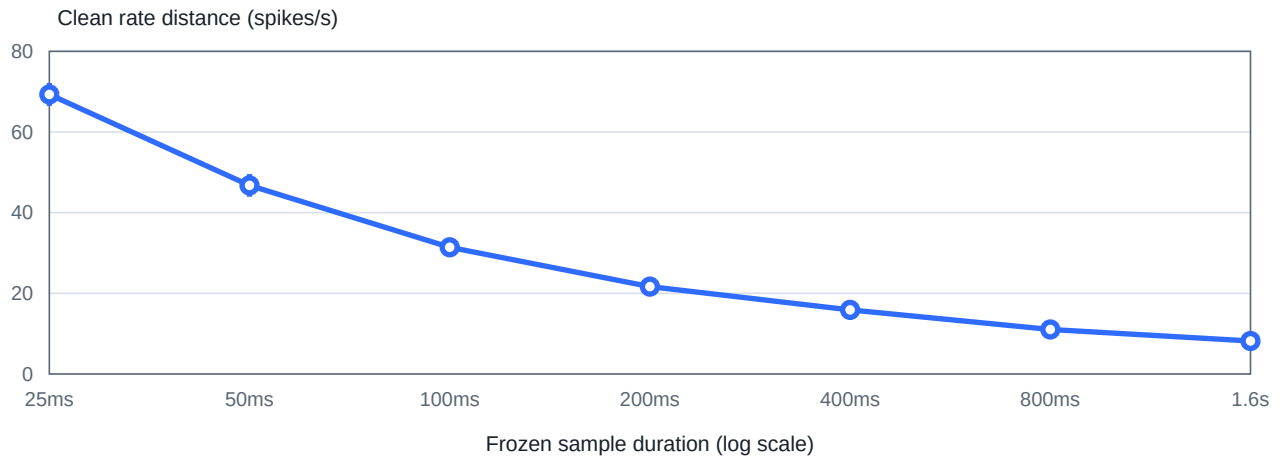


Figure 2. Root-mean-square clean rate-vector distance between the two frozen samples, averaged across 12 pairs. Error bars show ± 1 SEM. The distance falls from 69.3 spikes/s at 25 ms to 8.2 spikes/s at 1.6 s.

Reading the two figures together

- The encoder gives every finite frozen pair a distinct multichannel rate pattern.
- Whole-trace averaging reduces that clean class separation by roughly a factor of eight across the tested duration range.
- Pre-pooling spike variability can partly average out over time, but fixed post-pooling noise cannot.
- Once the clean fingerprint is comparable to the fixed readout-noise scale, held-out classification approaches chance.

The first simulation therefore supports the proposed mechanism **within a whole-trace rate-code model**. It does not yet establish that this is the mechanism used by human auditory cortex.

Why the clean condition stays perfect

Without post-pooling readout noise, repetitions from each class remain tightly clustered around a deterministic rate fingerprint. Even a small difference is sufficient for a linear classifier. The duration effect appears behaviorally only when the shrinking fingerprint must be read through irreducible noise or another source of finite precision.

What is already learned

Observation	Implication
Short samples remain robust under strong readout noise.	Their accidental cross-channel structure is large relative to the noise scale.
Long-sample accuracy changes steeply with readout noise.	The predicted duration boundary is not fixed; it depends on internal precision.
Clean distance declines smoothly with duration.	The effect is explained by convergence of finite-sample statistics rather than a hard duration switch.

6. Source-statistics extension

Complementary tasks

The extension asks whether the same pooled rate representation can reproduce the opposing duration effects reported for auditory textures. In the exemplar task, the classifier learns two fixed waveforms from the same noise family. In the source task, it trains on independent waveforms from two families and must generalize to previously unseen exemplars.

Four Gaussian power-law families were generated within the 200-7,000 Hz model band: white ($1/f^0$), light-pink ($1/f^{0.5}$), pink ($1/f^1$), and brown ($1/f^2$). The light-pink condition supplies a deliberately subtle source contrast; canonical white-pink and pink-brown contrasts test stronger spectral differences.

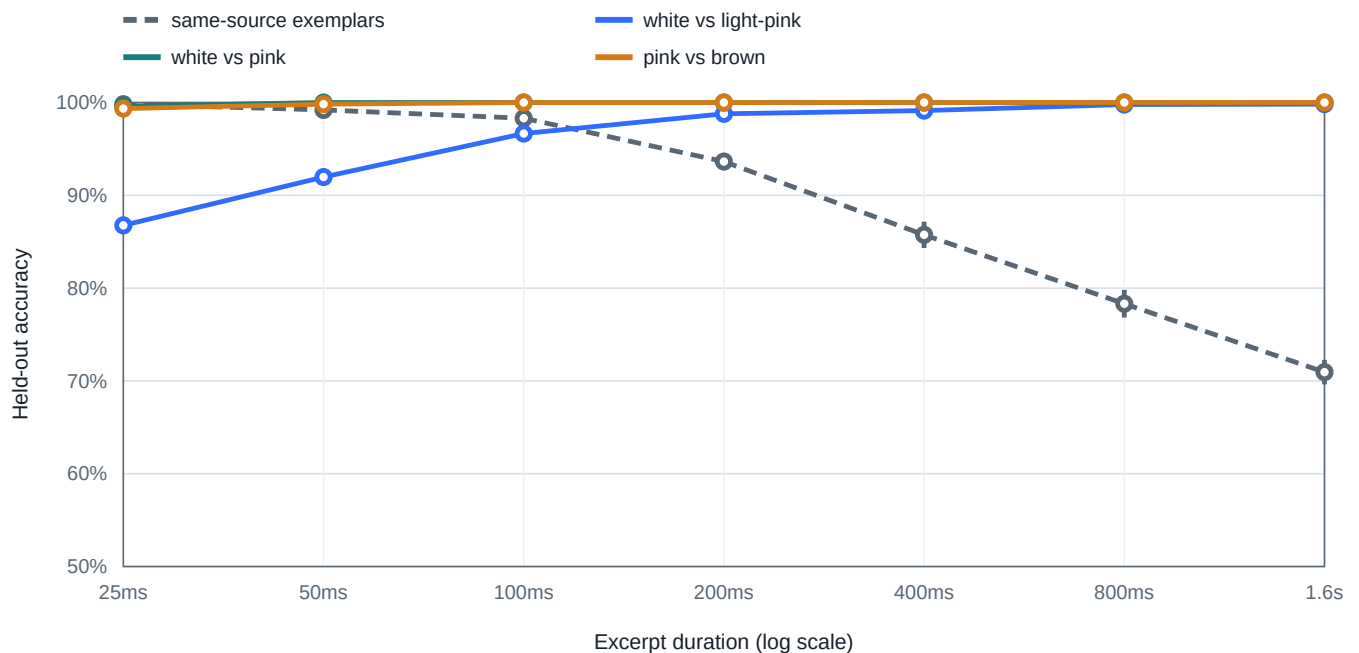


Figure 3. Accuracy at 20 spikes/s post-pooling readout noise. The dashed curve averages frozen-exemplar discrimination over all four families. Solid curves classify noise family from independent training and test exemplars. Error bars show ± 1 SEM across frozen pairs or train/test splits.

Duration	Same-source exemplars	White vs light-pink	White vs pink	Pink vs brown
25 ms	99.8%	86.8%	99.6%	99.3%
100 ms	98.3%	96.7%	100.0%	100.0%
200 ms	93.6%	98.8%	100.0%	100.0%
400 ms	85.7%	99.1%	100.0%	100.0%
1.6 s	70.9%	99.8%	100.0%	100.0%

Opposing duration effects. Same-source exemplar accuracy falls from 99.8% at 25 ms to 70.9% at 1.6 s. The deliberately difficult white versus light-pink classification rises from 86.8% to 99.8%, crossing the exemplar curve between 100 and 200 ms.

The mechanism is hierarchical: accidental exemplar deviations shrink with temporal averaging, but the stable mean spectral difference between sources remains. Canonical white-pink and pink-brown contrasts are nearly saturated even at 25 ms, showing that their source statistics are too different to reveal much duration dependence under these settings.

7. Limitations and next experiments

Important limitations

- The Fourier-domain filterbank approximates auditory frequency analysis but is not a causal gammatone or transmission-line cochlear model.
- Envelope threshold crossings omit adaptation, saturation, phase locking, multiple spontaneous-rate fiber classes, and correlated population noise.
- The whole-trace mean-rate decoder has no explicit memory limit and no invariance to temporal shifts or onset uncertainty.
- The readout-noise levels are expressed in model feature units and have not been calibrated to physiology or human psychophysics.
- Ceiling performance in the clean condition shows that finite classifier capacity alone is not currently limiting learning.

Highest-value next comparison

Train two matched decoders on the same spikes: (A) the current whole-trace rate code and (B) a time-resolved code that preserves spike counts in short bins. If only the rate decoder shows declining performance with duration, the result directly links the phenomenon to temporal information loss.

Suggested experiment sequence

Priority	Extension	Question answered
1	Rate code vs time-resolved code	Is temporal compression necessary for the duration effect?
2	Independent sweeps of dropout, spontaneous spikes, threshold jitter, and readout noise	Which noise location changes the duration boundary?
3	Leaky integration with a tunable time constant	Does a finite cortical memory window reproduce a sharper psychophysical transition?
4	Add acoustic noise before the filterbank	How does external SNR interact with frozen-sample duration?
5	Fit to behavioral data	Which parameters explain listener-level thresholds and individual differences?

Reproduce the current sweep

```
python3 -m venv .venv
.venv/bin/pip install -e .
frozen-noise-demo
PYTHONPATH=src python -m unittest discover -s tests
```

Conclusion

The model now reproduces both sides of the qualitative psychophysical result. Short frozen noises generate large accidental fingerprints across cochlear channels, but those fingerprints contract as duration grows. Stable differences between noise-source spectra persist and become easier to estimate as within-source fluctuations average away. The result is mechanistically interpretable, while its most important alternatives - preserved temporal coding and discrimination based on nonspectral texture statistics - remain straightforward to test next.

Appendix A. Where variability enters

A.1 From a stochastic waveform to a clean spike count

For source family s and exemplar e , the waveform $x_{(s,e)}(t)$ is sampled once and then frozen. Its power spectral density is controlled by a source-specific exponent:

$$\text{PSD}_s(f) \propto f^{-\alpha_s}$$

$$\alpha_s = 0 \text{ (white), } 0.5 \text{ (light-pink), } 1 \text{ (pink), or } 2 \text{ (brown)}$$

The waveform is filtered in cochlear channel k , converted to an analytic envelope, and thresholded. A spike is recorded only at an upward crossing, subject to the 1 ms refractory interval:

$$y_{(s,e,k)}(t) = (h_k * x_{(s,e)})(t)$$

$$a_{(s,e,k)}(t) = |y_{(s,e,k)}(t) + i H\{y_{(s,e,k)}\}(t)|$$

$$N_{(s,e,k)}(T) = \sum_{0 \leq t < T} 1[\text{upward crossing of threshold}]$$

The resulting clean rate separates into a stable source mean and an accidental finite-exemplar deviation:

$$N_{(s,e,k)}(T) / T = \mu_{(s,k)} + \delta_{(s,e,k)}(T)$$

$$\text{SD}[\delta_{(s,e,k)}(T)] \propto T^{-1/2}$$

Important distinction. The frozen waveform is the signal being classified. Its exemplar deviation is variability across different waveforms, not fresh noise added on every presentation.

A.2 Presentation noise before temporal pooling

On presentation j , every clean spike is independently retained with probability $1-p$. The detected count D therefore follows a binomial distribution:

$$D^{(j)}_{(s,e,k)} \sim \text{Binomial}(N_{(s,e,k)}, 1-p), \quad p = 0.03$$

$$\text{Var}[D_k / T] = N_k p(1-p) / T^2 \approx \rho_k p(1-p) / T$$

Independent spontaneous events are then added before pooling:

$$B^{(j)}_k \sim \text{Poisson}(\lambda T), \quad \lambda = 0.2 \text{ spikes/s/channel}$$

$$\text{Var}[B_k / T] = \lambda / T$$

$$r_{\text{tilde}}^{(j)}_{(s,e,k)} = (D^{(j)}_{(s,e,k)} + B^{(j)}_k) / T$$

Both dropout and spontaneous-count variability are upstream of the whole-trace average. Their variance in rate units therefore decreases approximately as $1/T$.

Appendix A. Noise after pooling and duration effects

A.3 Fixed post-pooling readout noise

After the entire trace has been compressed to one rate per cochlear channel, independent Gaussian readout noise is added. Unlike count noise, its standard deviation is specified directly in spikes/s and does not decrease with T :

```
eta^(j)_k ~ Normal(0, sigma_read^2)
r^(j)_(s,e,k) = max[0, r_tilde^(j)_(s,e,k) + eta^(j)_k]
sigma_read = 0, 10, 20, or 40 spikes/s
```

Complete stochastic feature equation

```
r^(j)_(s,e,k)(T) = max[0,
{Binomial(N_(s,e,k)(T), 0.97) + Poisson(0.2 T)} / T
+ Normal(0, sigma_read^2) ]
```

Ignoring the small nonlinearity introduced by clipping negative rates at zero, the conditional trial variance is approximately:

```
Var[r_k(T)] approx {rho_k p(1-p) + lambda} / T + sigma_read^2
```

Variability	Location	Changes across	Rate-variance scaling
Frozen-exemplar deviation	Stimulus, before filterbank	Independent waveforms	SD approx $T^{-1/2}$
Spike dropout	After spikes, before pooling	Presentations	Variance approx $1/T$
Spontaneous spikes	Before pooling	Presentations	Variance = λ/T
Gaussian readout	After pooling	Presentations	Variance = σ_{read}^2

A.4 Why the two duration effects oppose each other

For two exemplars of the same source, the stable source mean cancels. The distinguishing signal is the difference between two shrinking exemplar deviations:

```
Delta r_ex(T) = delta_(s,e1)(T) - delta_(s,e2)(T)
d'_ex(T) propto T^(-1/2) / sqrt(c/T + sigma_read^2)
```

For two different sources, the stable source separation remains while within-source exemplar variability averages away:

```
Delta r_source(T) = mu_s1 - mu_s2 + delta_(s1,e1)(T) - delta_(s2,e2)(T)
d'_source(T) propto ||mu_s1 - mu_s2|| / sqrt(c/T + sigma_read^2)
```

The exemplar signal shrinks toward a fixed readout-noise floor, so exemplar discrimination declines. The source signal remains fixed while within-source variability falls, so source discrimination improves and then plateaus.

A.5 Acoustic masking noise is not yet included

The present model does not add a fresh acoustic masker before the filterbank. A future external-noise condition would use $x_{\text{presented}}^{(j)}(t) = x_{\text{frozen}}(t) + \beta n^{(j)}(t)$.